

Mhamed Garbaa

Junior AI Engineer

📍 Tunisia ✉ mhamedgarbaa04@gmail.com 🔗 <https://mhamedgarbaa.github.io/portfolio/> 🔗 <https://linkedin.com/in/mhamed-garbaa-646509223>

Research Profile

Experience in applied AI and strong proficiency in Python, with hands-on expertise in developing and deploying intelligent solutions. Skilled in Generative AI, machine learning, AI agents, and data-driven systems, with a strong ability to translate complex concepts into practical applications. Proven autonomy in solving technical problems end-to-end, from design and development to deployment and optimization. Motivated to contribute to innovative projects by combining strong engineering skills, analytical thinking, and a research-oriented approach.

Professional Experience

AI Engineering Intern — Enterprise AI & Agentic Systems Talan Tunisia Feb. 2026 – Jun. 2026

- Designed and experimentally evaluated AI architectures for contextual reasoning and intelligent decision-making.
- Developed machine-learning and retrieval pipelines involving semantic representations, structured data, and neural models.
- Developed and experimentally evaluated retrieval, routing, and contextual reasoning mechanisms for intelligent workflows.
- Investigated how persistent context and structured knowledge can improve the reliability and autonomy of intelligent systems.
- Worked with Python, FastAPI, PyTorch-compatible AI ecosystems, graph-based representations, and large-scale model APIs.

AI Research Intern — Explainable Machine Learning Talan Tunisia Jun. 2025 – Aug. 2025

- Investigated Machine Learning and Explainable AI methods for data-driven decision systems.
- Designed experiments to evaluate models and explanation techniques based on **fidelity, interpretability, and consistency**.
- Explored **model adaptation and optimization** using open-source neural models and LoRA.

AI Research Intern — Multimodal Systems in Computational Biology Talan Tunisia Jul. 2024 – Aug. 2024

- Explored the application of **AI to computational biology** through a multimodal research project.
- Developed an interactive system for **3D protein structure visualization**, integrating SMfold with a Unity-based environment.
- Implemented **molecular toxicity prediction** components and documented the multimodal AI architecture.

Education

ENET'Com — National School of Electronics and Telecommunications of Sfax 2023 – 2026

Engineering Degree in **Telecommunications Engineering**, specialized in **Artificial Intelligence, Network Security**.

IPEIN, Preparatory Institute for Engineering Studies of Nabeul 2022 – 2023

Technical Skills

Programming: Python, PHP, SQL, Java, Bash, C/C++

Backend & APIs: FastAPI, Django REST, REST APIs, API Integration, Backend Development

Databases: PostgreSQL, MySQL, Neo4j, LanceDB, SQL, Vector Databases

AI & LLM: Generative AI, LLMs, AI Agents, Agentic AI, RAG, LangChain, LangGraph, Embeddings, Knowledge Graphs

ML: PyTorch, TensorFlow, Scikit-learn, Deep Learning, Neural Networks, Model Evaluation, Explainable AI

Engineering & DevOps: Git, Docker, Docker Compose, MLflow, CI/CD, GitLab, Jenkins, AWS, Azure

Certifications

- Microsoft AI Fundamentals (AI-900)
- Microsoft Azure Fundamentals (AZ-900)
- Machine Learning in Production – DeepLearning.ai
- MLOps Fundamentals – DataCamp
- Jenkins for CI/CD – KodeKloud
- Supervised Learning with Scikit-learn – DataCamp

[\[Link\]](#) 🔗

Competitions

- 3rd place – NASA Space Apps Challenge
- 3rd place – TomorrowLab (TPL / DAAD)

Extracurricular Activities

- President** – TUNSA Club
- Vice-President of Training and Development** – JCI Korba

Skills: Leadership, Strategic Planning, Teamwork

Languages

Arabic: Native **English:** C1 **French:** B2 (TCF-TP) **German:** A2